ENGINEERING PROJECT REPORT • ACADEMIC EXAMINATION & LABORATORY ARCHIVE

Autonomous Mobile Manipulator with 4-DOF Arm & Edge-AI with LiDAR Fusion for Pick-and-Place Tasks

An integrated mobile robotics platform combining differential-drive mobility, computer vision, LiDAR sensing, ROS 2, edge computing, and a DIY 4-DOF robotic manipulator.

> **Current Prototype Status:** Mobile base, LiDAR/camera ROS 2 infrastructure, remote motor control, and visualization are established. The 4-DOF arm is being integrated; pick-and-place is currently performed manually and is intended to become autonomous.

1. INTRODUCTION & SYSTEM BACKGROUND

Mobile robots provide locomotion and navigation across environments but lack physical manipulation capability. Conversely, stationary robotic arms execute dexterous manipulation but operate inside a restricted workspace. Integrating an articulated manipulator onto a mobile chassis creates a **mobile manipulator**, enabling autonomous transit, target identification, and physical interaction with ground-level objects.

This project extends an existing ROS 2-based autonomous soccer robot. The legacy platform verified high-speed differential mobility, computer vision, remote networking, and reactive obstacle avoidance. The current phase expands this platform by integrating a DIY 4-Degree-of-Freedom (4-DOF) arm and parallel gripper for pick-and-place functionality.

Primary Hardware Subsystems & Roles

- **Raspberry Pi:** Robot-side edge computing hub. Executes high-level decision algorithms, hosts the ROS 2 node graph, parses serial LiDAR packets, and interfaces with the camera.
- **Arduino Uno:** Low-level deterministic real-time microcontroller. Generates 500 Hz PWM motor driver signals, sets direction logic, and accumulates wheel encoder pulses.
- **L298N Motor Driver:** Dual H-bridge driver delivering variable 12V power to the drive motors.
- **Two Geared DC Motors with Encoders:** 12V JGA25-370 gearmotors providing differential mobility and wheel odometry.
- **YDLIDAR X4:** 360-degree 2D laser scanner providing radial distance arrays for mapping and obstacle clearance.
- **Raspberry Pi Camera V2:** Sony IMX219 optical sensor providing frames for ball detection and angular bearing tracking.
- **ROS 2 Jazzy:** Central robotics middleware facilitating inter-process messaging, lifecycle nodes, and coordinate frames.
- **DIY 4-DOF Arm & Gripper:** Articulated manipulator delivering positional reach and grasping capability.

Principles of Sensor Fusion & Edge-AI

Sensor Fusion: Monocular cameras deliver semantic awareness (target identification and angular direction), but lack direct metric depth. 2D LiDAR delivers millimeter-level spatial distance across 360 degrees, but lacks semantic target classification. Combining camera bearing with planar LiDAR range eliminates ambiguity, verifying target position and path clearance.

Edge-AI Execution: Sensor processing runs directly on the Raspberry Pi. Initial development uses color segmentation (HSV), with lightweight neural detectors (such as MobileNet-SSD or quantized YOLO models) planned for deployment without network latency.

Five Chronological Development Stages

Stage 1: Complete

Mobile Base & Sensor Setup

Chassis mobility, Arduino motor firmware, YDLIDAR raw packet parsing, and camera streaming.

Stage 2: Current Focus

4-DOF Arm Manual Integration

Arm mounting, power bus isolation, manual PWM testing, servo load verification, and grip testing.

Stage 3: Planned

Automated Ball Detection

OpenCV/Edge-AI vision node tracking target ball centroid and horizontal bearing angle.

Stage 4: Planned

Camera-LiDAR Fusion Approach

Correlating vision bearing with LiDAR range to navigate safely to grasping distance.

Stage 5: Final Target **Full Autonomous Pick-and-Place:** Autonomous state machine coordinating target search, navigation, kinematic arm descent, grasping, transit, and deposition.

2. PROJECT OBJECTIVES

> **Primary Objective**
>
> *"To develop an integrated mobile manipulator capable of detecting a ball/object, navigating toward it, and performing pick-and-place operations using a 4-DOF robotic arm."*

Specific Engineering Objectives

1. Build and operate a differential-drive mobile chassis with closed-loop encoder monitoring.
2. Deploy Raspberry Pi as the edge-computing, communication, and ROS 2 middleware master.
3. Stream real-time visual frames using the Raspberry Pi Camera Module V2 over CSI bus.
4. Acquire 360-degree planar range arrays using the YDLIDAR X4 laser scanner.
5. Implement ROS 2 Jazzy for modular, node-based inter-process communication.
6. Implement computer-vision/Edge-AI algorithms to extract target ball centroid coordinates.
7. Process LiDAR data to establish obstacle awareness and reactive navigation.
8. Fuse camera bearing with LiDAR distance measurements at the decision-making level.
9. Integrate a lightweight DIY 4-DOF articulated arm and gripper onto the mobile chassis.
10. Operate the arm manually to validate joint range, servo torque, and mechanical stability.
11. Progress toward kinematic modeling, calibrated workspace mapping, and automated trajectory execution.
12. Develop an end-to-end autonomous state machine for picking, transporting, and placing targets.

3. METHODOLOGY & WORKING PRINCIPLE

The platform operates on a closed-loop **Perception – Decision – Navigation – Manipulation** pipeline. Sensor data streams concurrently to edge compute nodes to generate mobile base and arm actuation commands.

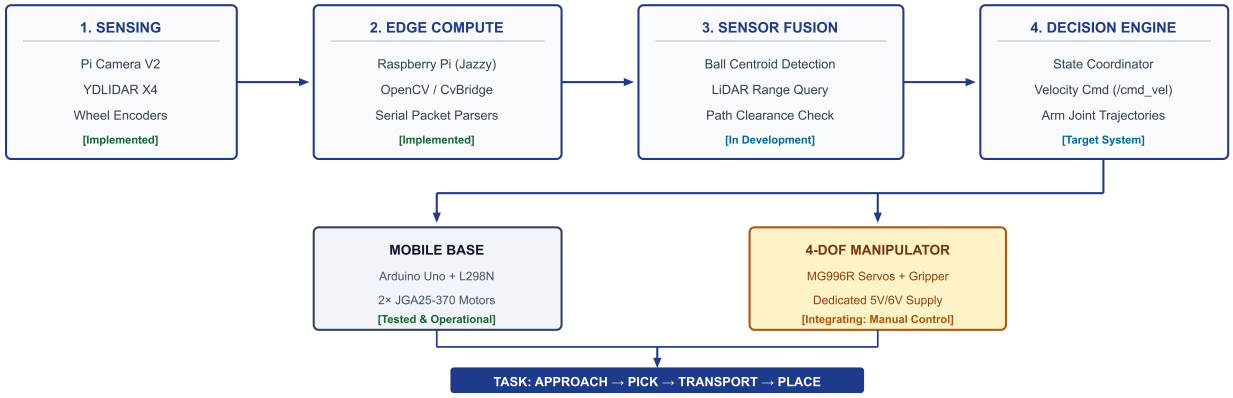


Figure 1: Complete System Hardware and Dataflow Block Diagram.
Current prototype: manual arm pick-and-place. Future target: automatic ball detection and autonomous manipulation.

3. METHODOLOGY (CONTINUED)

A. Core ROS 2 Middleware Topics

The software architecture runs on ROS 2 Jazzy Jalisco using the `rclpy` and `rclcpp` client libraries:

- `/scan` (`sensor_msgs/msg/LaserScan`): Transmits 360-degree range arrays from YDLIDAR X4.
- `/image_raw` (`sensor_msgs/msg/Image`): Streams raw RGB image frames from the Camera V2 node.
- `/cmd_vel` (`geometry_msgs/msg/Twist`): Carries linear and angular velocity commands for base locomotion.
- `/robot_description` (`std_msgs/msg/String`): Contains the unified URDF model describing base links and arm joints.
- `/tf` & `/tf_static`: Publishes real-time coordinate transformations relating `base_link`, `laser_frame`, `camera_link`, and `arm_base`.

B. Image Processing & Bearing Calculation

The vision pipeline converts raw BGR frames to the HSV color space to minimize sensitivity to ambient illumination changes. Morphological filtering (erosion and dilation) removes background noise, and spatial moments calculate the target centroid (u_c, v_c). The horizontal bearing angle θ relative to the robot optical center is calculated by:

$$\theta = \left(\frac{u_c - W/2}{W/2}\right) \times \left(\frac{\text{FOV}_h}{2}\right)$$

Where W is frame width (640 pixels) and FOV_h is horizontal field of view (62.2°).

C. LiDAR Range Association

The calculated bearing angle θ is indexed into the YDLIDAR scan array to retrieve metric target range:

$$k = \text{round}\left(\frac{\theta - \text{angle_min}}{\text{angle_increment}}\right), \quad r = \text{ranges}[k]$$

The target vector in Cartesian space relative to the robot base is $\mathbf{P} = [r\cos\theta, \; r\sin\theta]^T$.

D. Differential-Drive Kinematics

Forward kinematics transforms wheel angular velocities (ω_R, ω_L) into base linear velocity v and angular velocity ω:

$$v = \frac{R}{2}(\omega_R + \omega_L), \quad \omega = \frac{R}{L}(\omega_R - \omega_L)$$

Inverse kinematics computes required wheel velocities from commanded `/cmd_vel`:

$$\omega_R = \frac{v + (\omega \cdot L / 2)}{R}, \quad \omega_L = \frac{v - (\omega \cdot L / 2)}{R}$$

Where R is wheel radius and L is wheel track baseline.

E. DIY 4-DOF Robotic Arm Mechanics

The manipulator consists of four revolute joints plus a parallel-linkage gripper:

- **Base Yaw (Joint 1):** MG996R servo providing 180° horizontal panning.
- **Shoulder Pitch (Joint 2):** MG996R high-torque servo handling vertical lifting loads.
- **Elbow Pitch (Joint 3):** MG996R servo driving the forearm extension link.
- **Wrist Pitch (Joint 4):** SG90 micro-servo maintaining gripper ground parallelism.
- **Gripper End-Effector:** SG90 servo driving dual claws with silicone friction pads.

> **Manual Arm Testing Protocol (Current Development Stage)**
>
> The arm is currently operated manually via calibrated PWM signals. This validates: (1) mechanical stability without chassis tipping at maximum forward reach, (2) thermal stability of the shoulder servo under load, (3) electrical isolation of the 6V servo rail from the 5V logic bus, and (4) grasping retention on 50–150g spherical targets.

3. METHODOLOGY (CONTINUED)

F. Planned Autonomous Pick-and-Place Sequence (Target System)

The complete workflow follows a deterministic 16-step operational sequence:

Step	System Phase	Sensory / Algorithmic Trigger	Actuator / Hardware Action
1	**System Start**	ROS 2 nodes launched, TF initialized	Arm holds stowed home position; wheels stopped.
2	**Visual Search**	Pi Camera V2 active frame capture	Chassis rotates slowly in place.
3	**Ball Detection**	HSV color / Edge-AI contour match	Centroid pixel coordinates calculated.
4	**Bearing Estimation**	Optical geometry formulation	Horizontal bearing angle θ derived.
5	**LiDAR Range Query**	Scan indexed at bearing angle θ	Exact metric distance r retrieved; clearance checked.
6	**Chassis Alignment**	Visual/LiDAR bearing error feedback	Mobile base rotates to align θ to 0.
7	**Approach Navigation**	LiDAR obstacle avoidance monitoring	Base moves forward toward the ball.
8	**Stopping & Docking**	LiDAR range threshold reached ($r \approx 18\text{ cm}$)	Base halts at precise arm manipulation distance.
9	**Kinematic Solution**	Inverse kinematics solver execution	Target joint angles ($\theta_1, \theta_2, \theta_3, \theta_4$) calculated.
10	**Arm Descent**	Joint position trajectory interpolation	Arm links lower smoothly toward the ball.
11	**Object Grasping**	Gripper command execution	Claw closes securely around target ball.
12	**Payload Elevation**	Joint trajectory generation	Arm lifts the ball to carry height (~80 mm).
13	**Transit Navigation**	LiDAR navigation / pre-set waypoint	Chassis drives toward deposition receptacle.
14	**Receptacle Align**	LiDAR proximity / marker detection	Base halts; arm extends over container.
15	**Object Release**	Release trigger command	Gripper opens, depositing the ball.
16	**Return to Home**	Task complete notification	Arm returns to stowed position; awaits next command.

4. APPARATUS & TECHNICAL TOOLS

The hardware, software, and networking components used across the project:

Component / Tool	Primary Purpose & Technical Role
Raspberry Pi 3B	Edge computing hub; hosts ROS 2, executes vision algorithms, and manages communication.
Raspberry Pi Camera V2	8MP CSI sensor; provides optical video frames for ball detection.
YDLIDAR X4	360-degree laser rangefinder; provides 2D distance arrays for obstacle avoidance and ranging.
Arduino Uno	Low-level MCU; generates PWM motor signals and counts encoder pulses.
L298N Motor Driver	Dual H-bridge module delivering 12V drive current to the two DC motors.
2× Geared DC Motors	12V JGA25-370 gearmotors with Hall encoders providing differential locomotion.
DIY 4-DOF Robotic Arm	Articulated manipulator providing physical reaching and positioning.
Servo Motors	MG996R (base/shoulder/elbow) and SG90 (wrist/gripper) joint actuators.
Gripper	Two-finger mechanical claw with friction pads for ball grasping and release.
Chassis & Fittings	Differential-drive circular chassis, rubber wheels, hardware standoffs, and passive caster.
Battery & Power System	3S LiPo battery with separate buck regulators for logic (5V) and servos (6V).
Laptop / PC	Remote workstation used for development, ROS 2 monitoring, and RViz2 visualization.
ROS 2 Jazzy	Robotics middleware managing publish/subscribe communication and transform trees.
Python 3	Primary development language for ROS 2 nodes and algorithmic logic.
OpenCV	Computer vision library used for HSV conversion, filtering, and contour extraction.
NumPy	Performs vector operations and matrix calculations on sensor data.
CvBridge	Converts between ROS standard Image messages and OpenCV image matrices.
RViz2	3D visualization suite for real-time monitoring of LiDAR scans, TF trees, and URDF models.
Wi-Fi / TCP/IP / SSH	Wireless networking protocols enabling remote command, telemetry, and debugging.

5. APPLICATIONS & RESEARCH DIRECTIONS

The platform serves as an accessible testbed for academic research and robotics education:

- **Educational Robotics:** Instruction in distributed middleware, embedded controllers, and arm kinematics.
- **Mobile Manipulation Research:** Evaluating center-of-mass shifts, tip-over stability, and coordinated base-arm control.
- **Warehouse Logistics Prototyping:** Small-scale research into automated parcel retrieval, sorting, and transport.
- **Laboratory Sample Handling:** Autonomous transport of specimen vials between diagnostic processing stations.
- **Robotic Competitions:** Baseline architecture for autonomous retrieval tasks and RoboCup Logistics trials.
- **Edge-AI Benchmarking:** Profiling quantized neural detection models on resource-constrained single-board computers.
- **Sensor Fusion Studies:** Testing 2D spatial projection methods linking camera pixels with laser ranges.
- **Human-Robot Interaction (HRI):** Investigating collaborative handover routines and shared workspace protocols.
- **Service Robotics:** Prototyping indoor assistance tasks such as floor clearing and object retrieval.
- **Flexible Workcells:** Small-scale component transfer in modular automated workcells.

Note: The platform is designed as an academic research and prototyping testbed, not for direct heavy industrial deployment.

6. PROJECT BUDGET (BANGLADESHI TAKA - BDT)

Estimated prototype component costs based on local supplier sourcing in Bangladesh:

Component Description	Estimated Cost (BDT)
Raspberry Pi / Single-Board Computing Unit	8,000 – 12,000
Raspberry Pi Camera Module V2 (8MP IMX219)	2,500 – 4,000
YDLIDAR X4 360° Laser Rangefinder	10,000 – 15,000
Arduino Uno R3 Microcontroller Board	800 – 1,500
L298N Dual H-Bridge Motor Driver Module	250 – 500
2 × Geared DC Motors with Integrated Encoders	2,500 – 4,000
DIY 4-DOF Robotic Arm Mechanical Linkage Kit	3,000 – 6,000
Servo Motors (High-Torque MG996R & SG90 set)	2,000 – 4,000
End-Effector Mechanical Gripper Claw	800 – 1,500
Chassis Plates, Wheels, Couplers, & Mechanical Fittings	2,000 – 4,000
Battery, Voltage Regulators, & Wiring Harness	2,000 – 4,000
Miscellaneous Fabrication, Standoffs, & Fasteners	1,000 – 2,500
Total Estimated Prototype Budget	**34,850 – 59,000 BDT**

Budget Clarifications & Incremental Cost

Actual costs depend on local suppliers, component availability, shipping, and existing hardware inventory. Because this project extends an existing ROS 2 soccer robot, major capital items (Raspberry Pi, YDLIDAR, motors, chassis, and Arduino) were already available. **The incremental cost of adding the 4-DOF arm, servos, gripper, and dedicated voltage regulators is approximately 8,000 to 15,500 BDT.**

7. CONCLUSION

This project presents the design, integration, and phased implementation of an autonomous mobile manipulator by combining a verified differential-drive mobile robot with a DIY 4-DOF articulated arm. The architecture unifies mobile robotics, robotic manipulation, ROS 2 middleware, edge computing, computer vision, planar LiDAR range sensing, sensor fusion, differential kinematics, embedded systems, and wireless networking.

The mobile chassis, low-level motor drives, sensor pipelines, and visualization tools are fully operational. At the current prototype stage, the 4-DOF arm is operated manually for pick-and-place testing. This systematic testing validates mechanical stability, servo torque, and grip security before closing the loop with automated kinematic control. The final project goal is to fully automate the complete workflow, from ball detection and navigation to grasping, transportation, and placement.

> **Key Engineering Takeaways**
>
> 1. Separating high-level edge computation (Raspberry Pi) from real-time motor control (Arduino Uno) ensures reliable performance without timing conflicts.
> 2. Fusing camera bearing with planar LiDAR range provides accurate metric localization of targets while maintaining obstacle awareness.
> 3. The platform serves as a modular, low-cost research and educational testbed for autonomous mobile manipulation.

8. TEN-POINT FUTURE DEVELOPMENT ROADMAP

The following roadmap outlines the technical milestones required to transition from manual arm validation to full autonomy:

1. Camera-LiDAR Calibration: Compute the extrinsic transformation matrix aligning camera optical frames with laser coordinates.	**2. Robust Ball Detection:** Deploy a lightweight Edge-AI model (e.g., MobileNet-SSD via TFLite) for robust multi-lighting detection.
3. Real-Time Sensor Fusion: Automate bearing-to-range coordinate projection for target tracking.	**4. Coordinate Frame Calibration:** Expand the URDF tree to include all arm joint transforms in ROS 2 TF.
5. Inverse Kinematics (IK): Implement analytical geometric IK equations to solve joint angles for target 3D positions.	**6. Automatic Gripper Control:** Automate grasping and release triggers based on end-effector proximity.
7. Visual Servoing: Implement closed-loop visual feedback during final approach to minimize docking errors.	**8. Odometry Improvement:** Fuse wheel encoder odometry with IMU data using an Extended Kalman Filter (EKF).
9. Unified State Machine: Implement a supervisory finite state machine using BehaviorTree.CPP.	**10. Experimental Benchmarking:** Conduct quantitative trials to benchmark pick-and-place success rates and cycle times.